

Project Documentation - Personal Portfolio & Management Platform

1. Project Overview

1.1 Purpose

The purpose of this project is to build a **full-stack personal portfolio platform** with a powerful **admin management system**.

It allows individuals to showcase:

- Their projects
- Professional skills
- Employment history
- Clients
- Analytics & trends
- AI & live chat interaction

At the same time, it provides administrators with full control over content, security, and UI settings.

This project is designed as a **real-world enterprise learning system** using modern technologies and best development practices

2. Technology Stack

2.1 Backend

- Java 11+
- Spring Boot
- Spring Data JPA
- Spring Security
- JWT Authentication
- Maven
- PostgreSQL / MySQL

2.2 Frontend – Client (Portfolio)

- React
- TypeScript
- Tailwind CSS
- Bootstrap
- Framer Motion
- Chart.js / Recharts

2.3 Frontend – Admin Panel

- Angular
 - TypeScript
 - Bootstrap
 - Chart.js
-

3. Core System Entities

Entity	Description
User	User account & profile management
Role	Permission & access control
Project	Portfolio project details
Client	Client information
PastEmployment	Employment history

Entity	Description
Skill	Skill and proficiency levels
UISettings	UI customization preferences
ChatMessage	AI & live chat messages
Payment	Demo payment records

4. System Functional Requirements

4.1 Portfolio Projects Module

Features

- Display projects as cards and carousels
- Admin can **add, edit, delete** projects
- Filter projects by skill or client

DSA Used

- ArrayList
 - Sorting (Comparator)
 - Searching
-

4.2 Clients Showcase Module

Features

- Display client logos in carousel format
- Associate clients with projects

DSA Used

- HashMap
 - Searching
-

4.3 Past Employment Timeline

Features

- Chronological employment history
- Animated timeline using Framer Motion

DSA Used

- LinkedList
 - Sorting
-

4.4 Skills & Highlights Module

Features

- Skill progress indicators
- Skill usage across projects

DSA Used

- Tree
 - Recursion
 - HashMap
-

4.5 AI & Live Chat Module

Features

- AI chatbot answers portfolio questions
- Real-time live chat between visitor and admin
- Message animations and typing indicators

DSA Used

- Queue
 - Stack
 - HashMap
-

4.6 Charts & Analytics Dashboard

Features

- Skill activity trends
- Project & client statistics
- API usage charts

DSA Used

- Prefix Sum
 - Sliding Window
 - Sorting
-

4.7 UI Customization Module

Features

- Light / Dark theme
- Enable / disable animations
- Layout customization
- Navbar, Sidebar, Footer, Button styling

DSA Used

- HashMap
 - Caching logic
-

4.8 Authentication, Security & RBAC

Features

- JWT-based authentication
- Role-based access control
- Protected admin APIs
- Secure password hashing

Security Concepts

- Hashing
 - Token validation
 - Role mapping
-

5. Data Structures & Algorithms Requirements

Each developer must implement **at least 7** of the following:

- Arrays / Lists
 - Stack
 - Queue
 - HashMap
 - Tree
 - Graph
 - Sorting
 - Searching
 - Greedy Algorithms
 - Recursion
 - Prefix Sum
 - Sliding Window
-

6. Testing & Quality Assurance Requirements

- JUnit tests
 - Mockito for mocking
 - Minimum **60% code coverage**
 - JaCoCo report generation
-

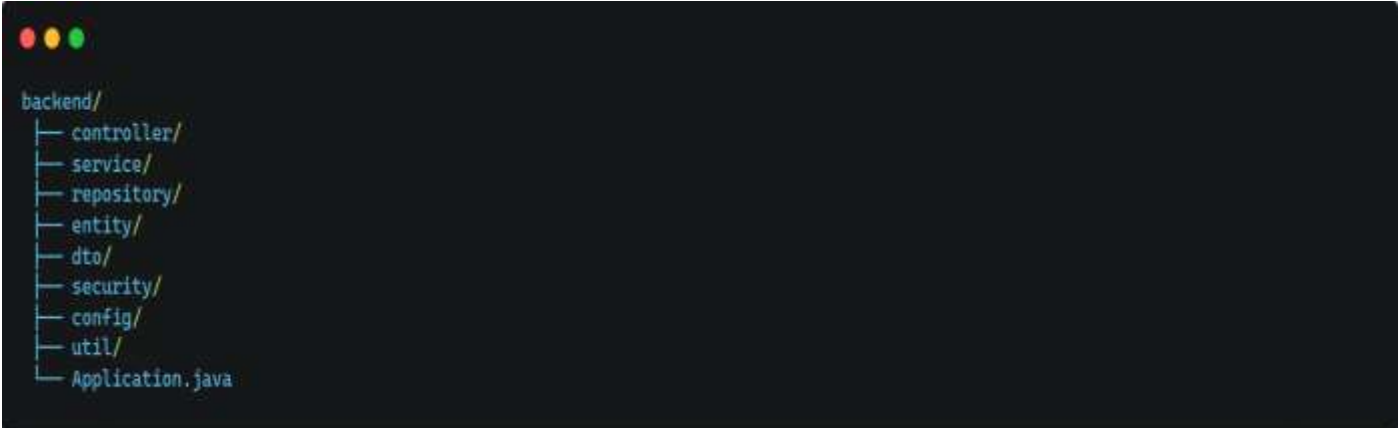
7. Documentation Requirements

The project must include:

- README.md
 - API_DOC.md
 - TESTING.md
 - SECURITY.md
 - DSA_EXPLANATION.md
-

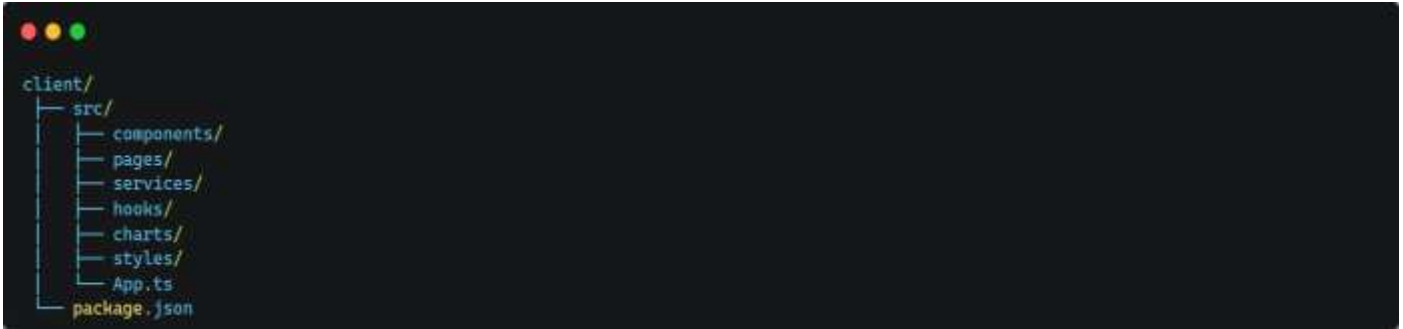
8. Project Structure

Backend (Spring Boot)



```
backend/  
├── controller/  
├── service/  
├── repository/  
├── entity/  
├── dto/  
├── security/  
├── config/  
├── util/  
└── Application.java
```

Frontend – Client (React)



Frontend – Admin (Angular)



9. Agile Development Plan

The project follows the **Agile Scrum methodology**.

Development is divided into **6 sprints**, each delivering a stable and testable product increment.

Sprint Rules

- Sprint Duration: **1 Week**
- Total Sprints: **6**
- Daily Code Commit Required
- Sprint Demo & Approval required before moving forward

Sprint 1 – Core Setup & Architecture Foundation

Objectives

Establish complete development environment and system foundation.

Requirements

- Create Spring Boot backend project
- Configure database (PostgreSQL / MySQL)
- Setup React client application
- Setup Angular admin application
- Configure Bootstrap, Tailwind CSS
- Create base project structure for backend and frontend
- Setup Git repository and branching strategy

Expected Outcomes

- All applications compile and run successfully
 - Base folder structures created
 - Database connectivity established
 - Initial UI layout visible on client and admin side
-

Sprint 2 – Portfolio, Clients & Employment Management

Objectives

Build the core portfolio content management.

Requirements

- Implement Project entity, repository, service, controller
- Implement Client entity and relationships
- Implement PastEmployment entity
- CRUD APIs for projects, clients, employment
- Display projects as cards on client UI
- Display clients in carousel format
- Display employment timeline
- Implement the Get In Touch Page.

Expected Outcomes

- Admin can manage projects, clients, employment
 - Portfolio displays real content dynamically
 - Data is stored and retrieved from database
-

Sprint 3 – Skills Dashboard & UI Customization

Objectives

Create skills showcase and UI management features.

Requirements

- Implement Skill entity and APIs
- Skill progress dashboard with charts (Chart.js)
- Skill-to-project mapping
- Implement UI Settings module:
 - Theme selection (Light/Dark)
 - Layout options
 - Navbar, Sidebar, Footer management
 - Button & color customization

Expected Outcomes

- Skills displayed with progress indicators
 - Analytics charts operational
 - UI can be customized from admin panel
 - Changes reflect instantly on client UI
-

Sprint 4 – Authentication, Security & RBAC

Objectives

Secure the platform and control access.

Requirements

- Implement User & Role entities
- JWT authentication flow
- Secure all admin APIs
- Role-based API authorization
- Angular route guards
- Password hashing & token validation
- Implement System Log, Application Tracing page

Expected Outcomes

- Admin panel accessible only after login

- Different roles enforce permissions
 - All protected APIs secured
 - System meets security standards
-

Sprint 5 – Chat System, Analytics & API Insights

Objectives

Add intelligent interaction and system analytics.

Requirements

- Implement AI chatbot (portfolio Q&A)
- Implement live chat between user and admin
- Real-time message handling (WebSocket or polling)
- Create ChatMessage entity and APIs
- Build analytics dashboard:
 - Skill trends
 - Project & client statistics
 - API usage metrics

Expected Outcomes

- Users can interact via AI & live chat
 - Admin can respond in real-time
 - Analytics visualized clearly
 - Message system stable and ordered
-

Sprint 6 – Testing, Hardening & Final Delivery

Objectives

Ensure system stability, performance, and documentation quality.

Requirements

- Write Junit & Mockito tests
- Achieve minimum **60% code coverage**
- Generate JaCoCo reports
- Finalize documentation:
 - README.md
 - API_DOC.md
 - SECURITY.md
 - TESTING.md
 - DSA_EXPLANATION.md
- Bug fixing & performance optimization
- Final project demo

Expected Outcomes

- Stable production-ready system
 - Verified code quality
 - Complete documentation set
 - Final approved release
-

10. Sprint Wise Evaluation Marks Structure

Total Marks 100

10.1 Sprint 1 Core Setup and Foundation

Evaluation Component	Marks
Project setup and build configuration	3
UI base layout Landing About Admin Login	3
Routing and navigation	2
Code structure and Git commits	2
Total	10

10.2 Sprint 2 Portfolio Clients and Employment

Evaluation Component	Marks
Project Client and Employment APIs	6
Dynamic UI rendering	6
Data persistence and database design	4
UI and UX quality	4
Total	20

10.3 Sprint 3 Skills Dashboard and UI Customization

Evaluation Component	Marks
Skill APIs and data mapping	5
Charts and visual analytics	5
UI management module	5
System integration quality	5
Total	20

10.4 Sprint 4 Authentication and Security

Evaluation Component	Marks
JWT authentication flow	5
RBAC implementation	4
Route and API security	4
Security best practices	2
Total	15

10.5 Sprint 5 Chat Analytics and API Insights

Evaluation Component	Marks
AI chat and live chat implementation	6
Analytics dashboard functionality	6
API metrics and visualization	4
System stability and performance	4
Total	20

10.6 Sprint 6 Testing Polish and Final Delivery

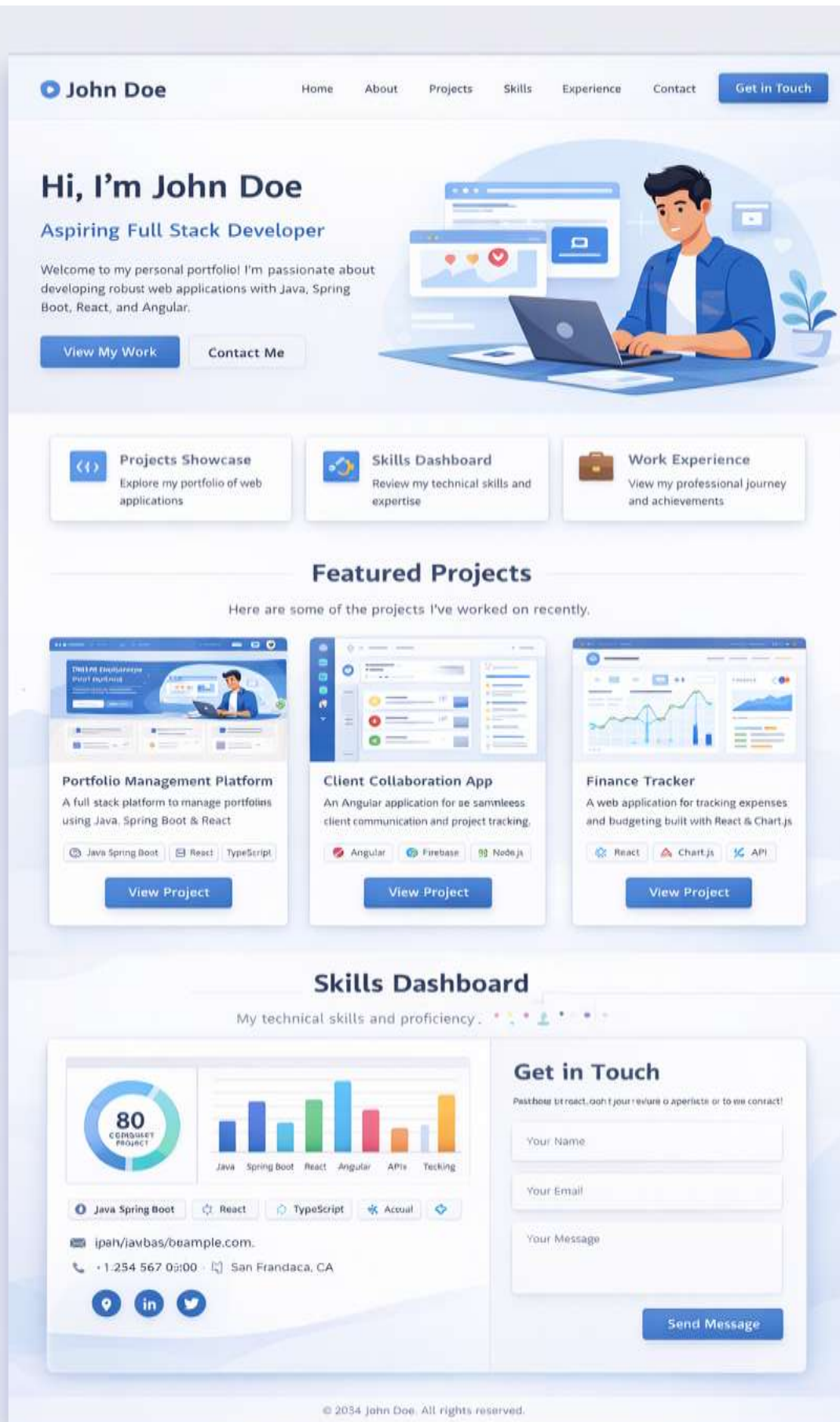
Evaluation Component	Marks
Automated testing and code coverage	6
Bug fixing and optimization	4
Documentation completeness	3
Final demo and presentation	2
Total	15

10.7 Final Marks Distribution Summary

Sprint	Marks
Sprint 1	10
Sprint 2	20
Sprint 3	20
Sprint 4	15
Sprint 5	20
Sprint 6	15
Total	100

11. Sprint Mockup Pages

I. Client Side – Home Page



Projects Showcase

Explore my portfolio of web applications



Skills Dashboard

Review my technical skills and expertise



Work Experience

View my professional journey and achievements

Featured Projects

Here are some of the projects I've worked on recently.



Portfolio Management Platform

A full stack platform to manage portfolios using Java, Spring Boot & React

Java Spring Boot React TypeScript

[View Project](#)

Client Collaboration App

An Angular application for seamless client communication and project tracking.

Angular Firebase Node.js

[View Project](#)

Finance Tracker

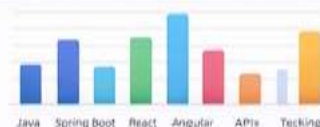
A web application for tracking expenses and budgeting built with React & Chart.js

React Chart.js API

[View Project](#)

Skills Dashboard

My technical skills and proficiency



Java Spring Boot React TypeScript Angular

ipah/javbas@example.com

+1 254 567 0300 San Francisco, CA

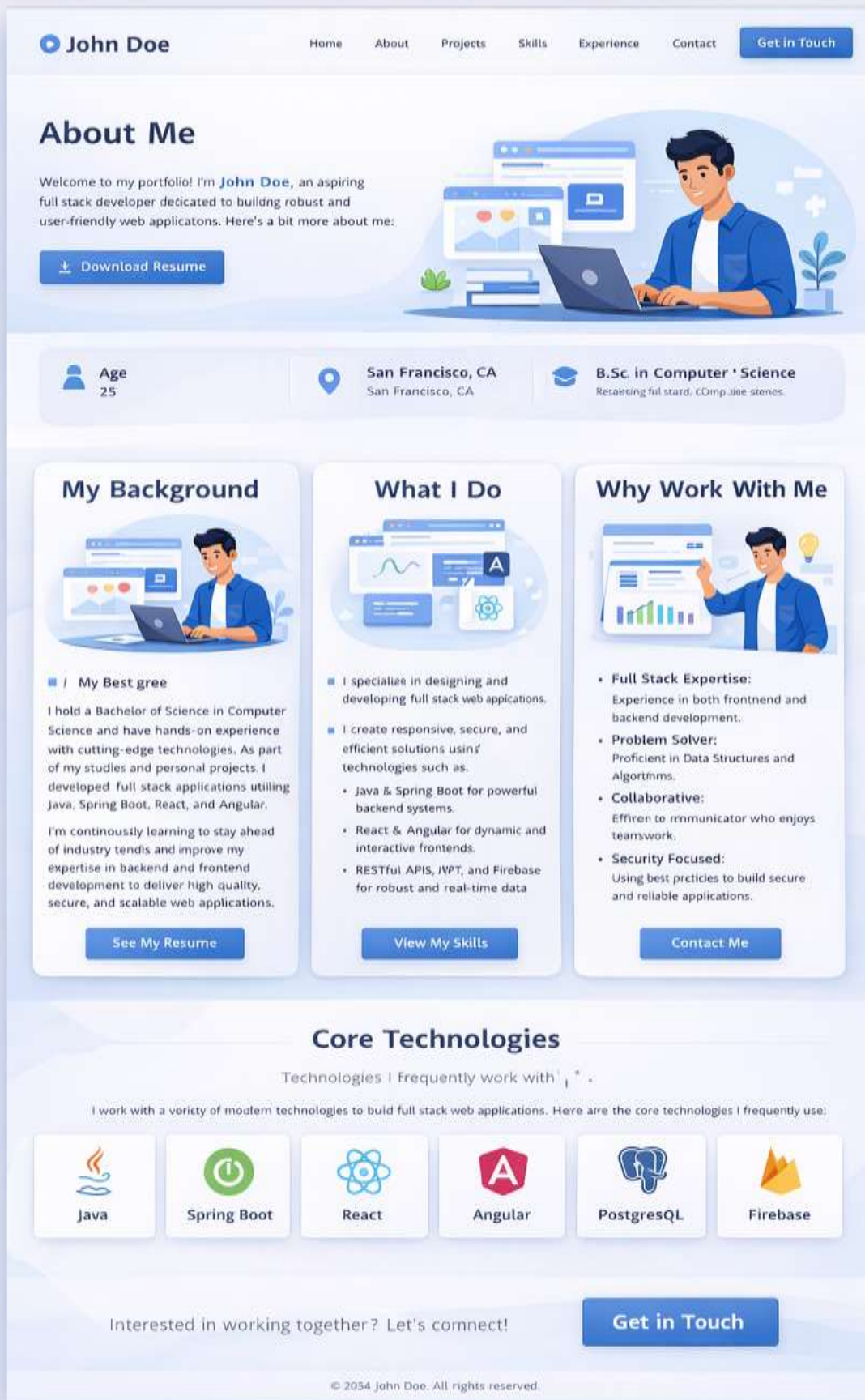


Get in Touch

Post here to reach out to your future employer or to me contact!

[Send Message](#)

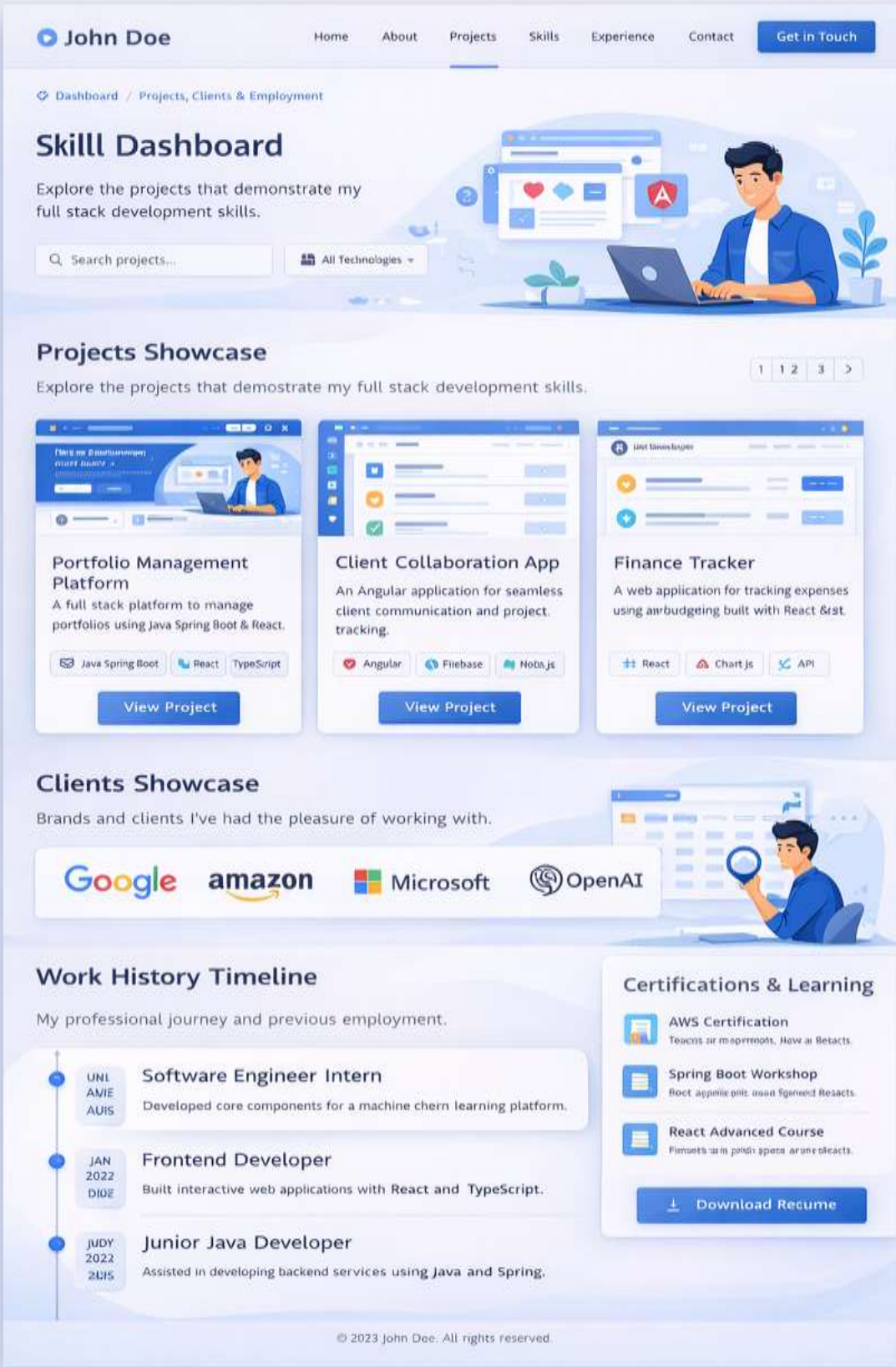
II. Client Side – About Page



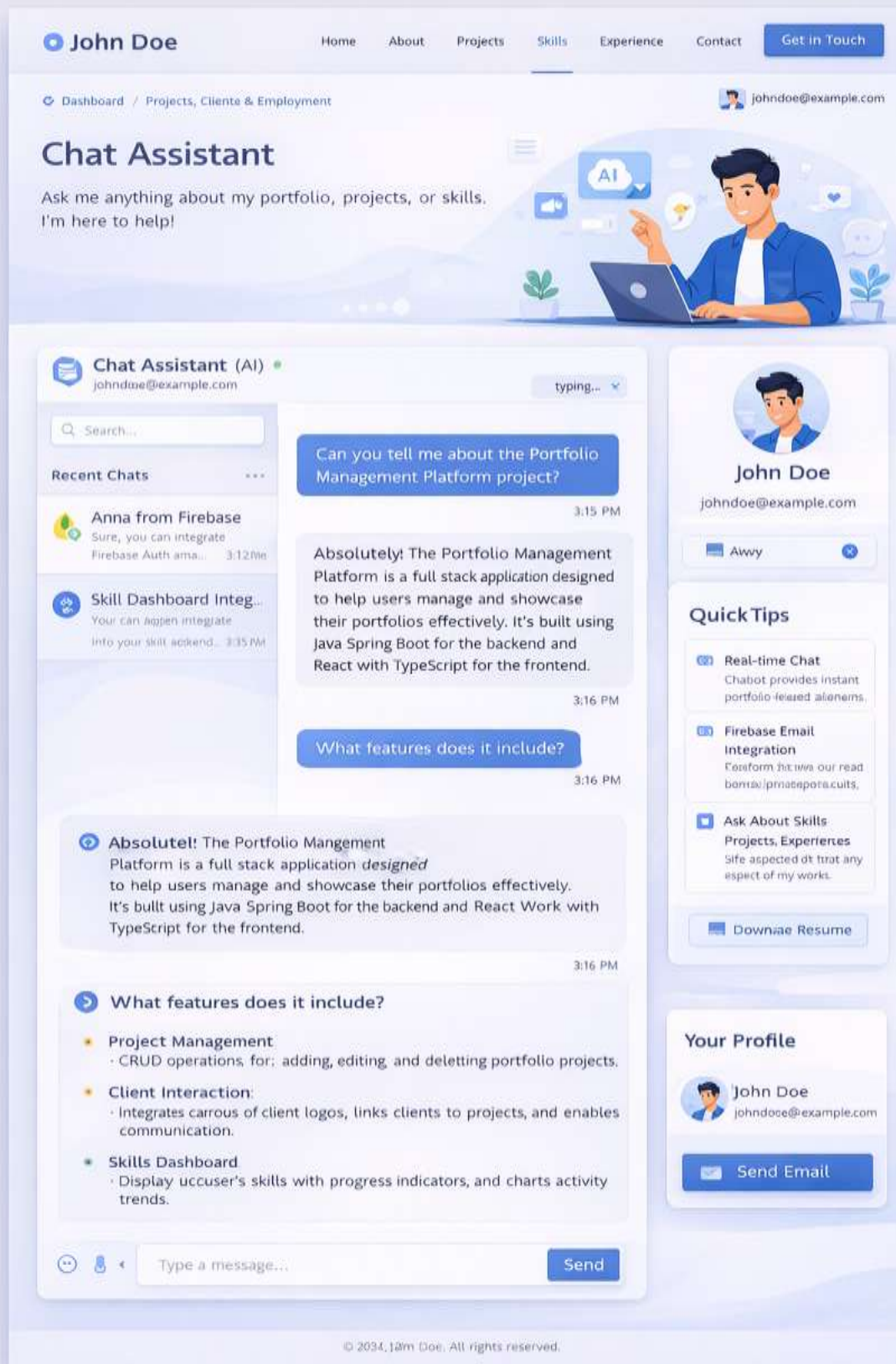
III. Client Side – Skill Page



IV. Client Side – Employment and Projects Page



V. Client Side - Chat Page



VI. Client Side – Get In Touch Page

 Home

 About

 Projects

 Skills

 Contact Me

Let's Chat!

Get in Touch

Feel free to reach out for project inquiries, collaborations, or just to say hello.



 Contact Information

Evaluation Component

 +123 456 7890

 someone@example.com

 1234 Developer Lane,
Tech City, 56789



 Send a Message

Full Name

 Full Name 

Email

 Your Email 

Subject

 Select Subject 

Message

Your Message

Send Message

 Thank you! Your message has been sent.

☐ I'm not a robot

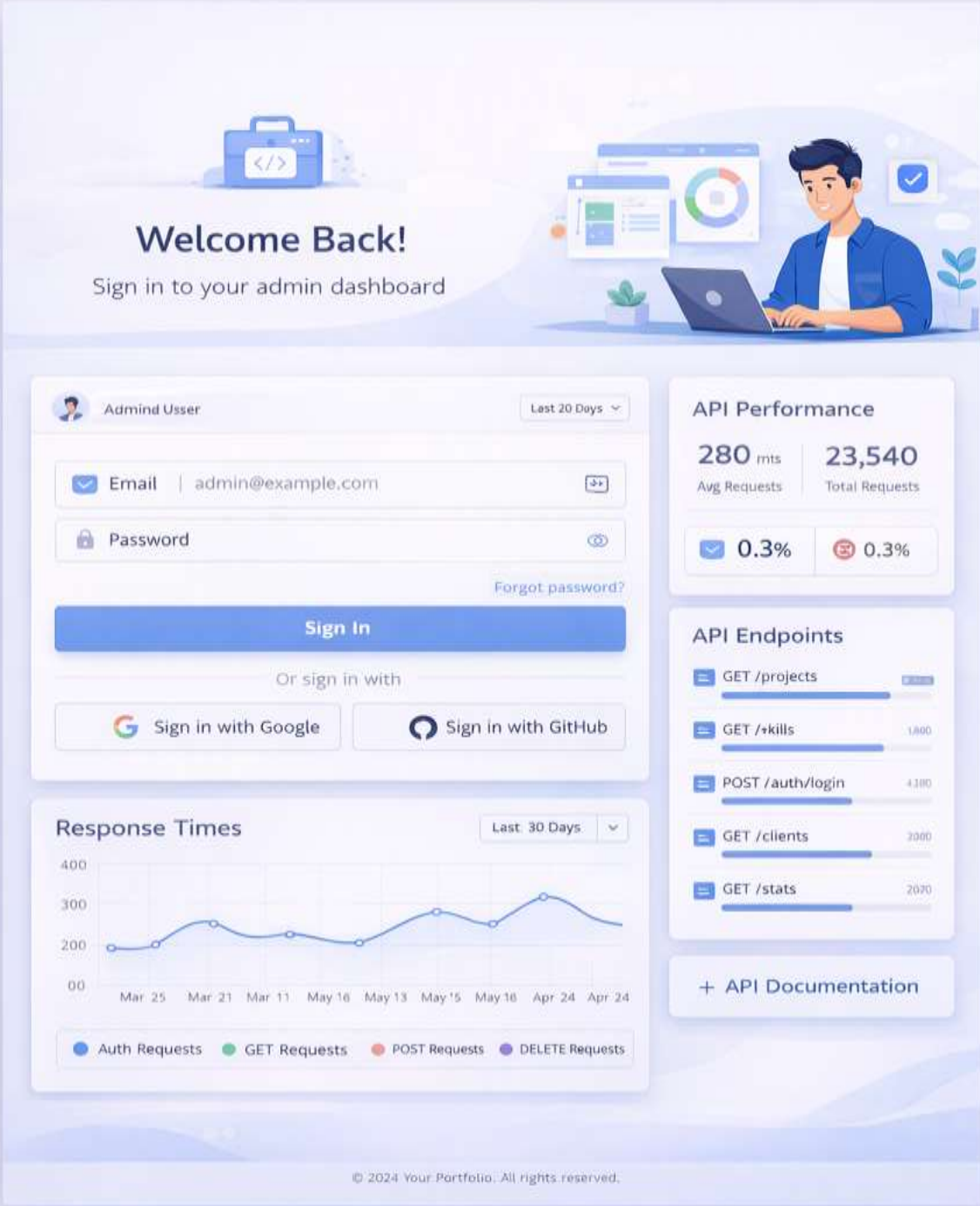


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VII. Client Side – API Analytics Page



VIII. Admin Side – Login Page



IX. Admin Side - Admin Dashboard

Admin Dashboard

admin@example.com

Toggle Theme

Dashboard

Skills

Employment

Projects

Clients

API Analytics

Employment

Projects

Clients

Project

Clients

Client

Admin Dashboard

Personal Portfolio Management

Skill Section

+ Add Skill

Search...

Search

Skill	Level	Proficiency	Actions	
JavaScript	90%	90%	Edit	Delete
React	85%	95%	Edit	Delete
Node.js	80%	80%	Edit	Delete
TypeScript	73%	75%	Edit	Delete
Python	78%	75%	Edit	Delete
SQL	70%	70%	Edit	Delete
AWS	65%	80%	Edit	Delete
Docker	60%	60%	Edit	Delete

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Employment Section

+ Add Employment

Search...

Search

Company	Title	Duration	Actions	
TechCorp	Senior Software Engineer	2019 - Present	Edit	Delete
WebSolutions	Frontend Developer	2017 - 2019	Edit	Delete
InnovateTech	Software Intern	2016	Edit	Delete

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Project Section

+ Add Project

Search...

Search

Client	Project Name	Tech Stack	Actions	
LogiSync	Portfolio Management Platform	React, Spring Boot, Chart.js, Framer	Edit	Delete
WebMetrics	E-commerce Website	React, Node.js, MongoDB, Svelte	Edit	Delete
StartupTech	Blogging Platform	TypeScript, Node.js, Next.js	Edit	Delete

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Client Section

+ Add Client

Client Name	Industry	Projects	Actions	
LogiSync	Software	6	Edit	Delete
WebMetrics	E-commerce	3	Edit	Delete
StartupTech	Saas	1	Edit	Delete

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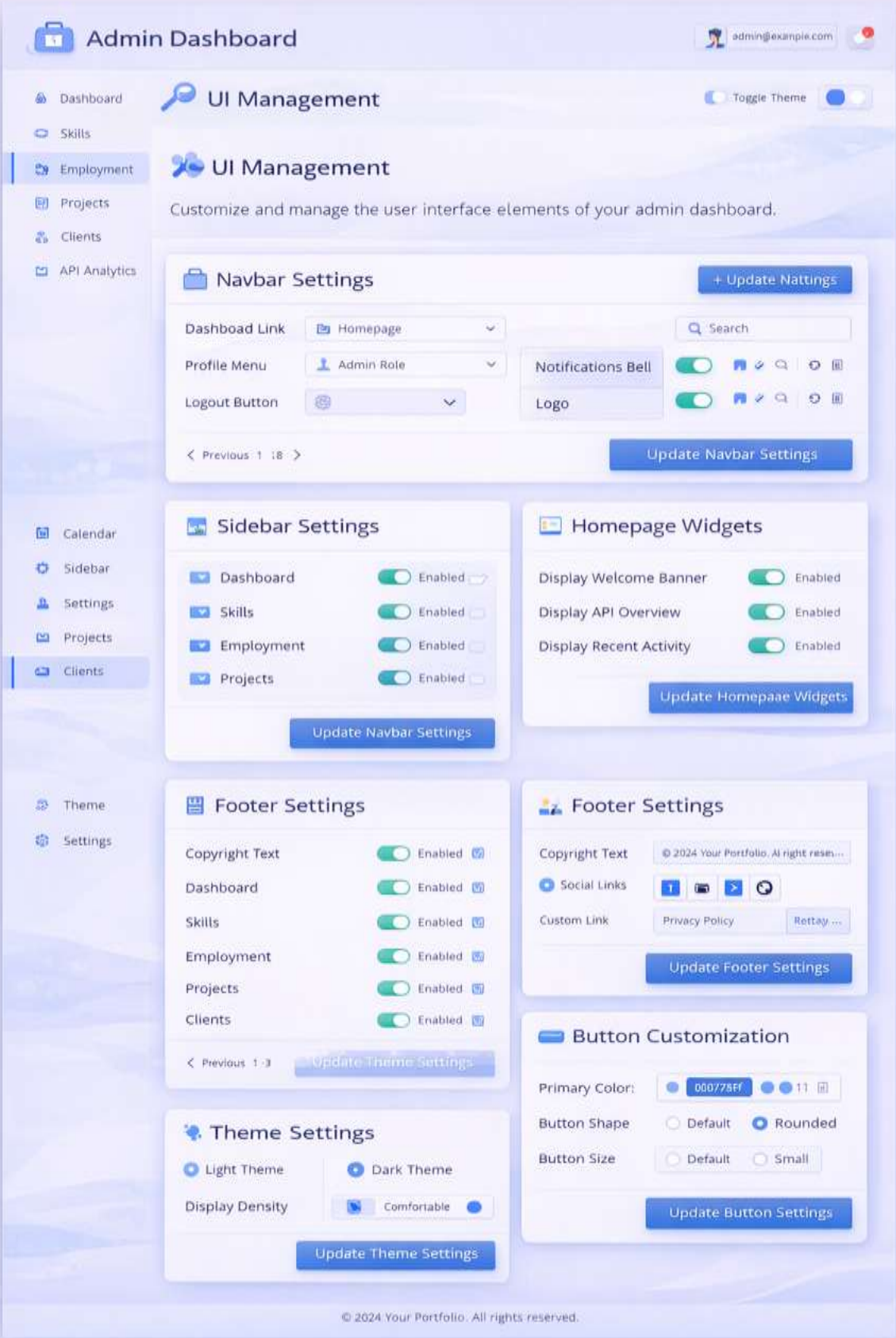
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X. Admin Side – UI Management



XI. Admin Side – Activity Monitoring, System Log Page

System Logs and Activity Monitor

Track system events, user actions, API activity, and application health in real time.

Total Requests

24,512

Active Users

124

System Errors

3

API Calls

18,950

Activity Logs Table

Admin

Export Logs

Clear Logs

Date Range

Today

Module

All Modules

Status

All Statuses

Search

Auto Refresh

Timestamp	User	Module	Action	Status	IP Address	
14 Jan 12:35:00 PM	admin_01	Portfolio	Update Project	Success	192.168.1.10	Details
14 Jan 12:35:00 PM	admin_02	Portfolio	Update Project	Success	192.168.1.10	Details
14 Jan 12:35:30 PM	admin_03	Portfolio	Update Project	Success	192.168.1.10	Details
14 Jan 12:35:30 PM	admin_04	Portfolio	Update Project	Success	192.168.1.10	Details
14 Jan 12:35:10 PM	admin_05	Portfolio	Update Project	Success	192.168.1.10	Details
14 Jan 12:35:20 PM	admin_06	Portfolio	Update Project	Success	192.168.1.10	Details
14 Jan 12:35:50 PM	admin_07	Portfolio	Update Project	Success	192.168.1.10	Details

API Request Logs

Endpoint	Method	Response Time
/api/projects	GET	120ms

Status Code	Status Code
200 OK	frontend_app

Triggered By	Timestamp
frontend_app	14 Jan 12:30:J, 12:55:03 PM

System Health Panel

Server Status

Healthy

Database Status

Connected

Error Rate

0.01%

Uptime

99.99%

Memory Usage

CPU Usage

Alerts and Events Panel

Critical

Database Connection Timeout

14 Jan 12:30 PM

Warning

High API Latency

14 Jan 11:45 AM

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USER			
Long	id	PK	
String	username		
String	email		
String	password_hash		
String	full_name		
String	bio		
String	profile_image_url		
String	contact_number		
LocalDateTime	created_at		

ROLE			
Long	id	PK	
String	role_name		e.g., ROLE_ADMIN, ROLE_USER
String	description		

CLIENT			
Long	id	PK	
String	name		
String	logo_url		
String	website_url		
String	description		

PAST_EMPLOYMENT			
Long	id	PK	
Long	user_id	FK	
String	company_name		
String	job_role		
LocalDate	start_date		
LocalDate	end_date		
Boolean	is_current		
String	description		Achievements

CHAT_MESSAGE			
Long	id	PK	
String	session_id	For guests	
Long	user_id	FK	Nullable
String	message_content		
String	sender_type		USER, ADMIN, AI
LocalDateTime	timestamp		
Boolean	is_read		

PAYMENT			
Long	id	PK	
Long	user_id	FK	Optional if guest
BigDecimal	amount		
String	currency		
String	transaction_id		
String	status		SUCCESS, FAILED, PENDING
String	payment_method		
LocalDateTime	timestamp		

UI_SETTINGS			
Long	id	PK	
Long	user_id	FK	
Boolean	dark_mode		Theme Selection
Boolean	animations_enabled		
String	layout_preference		
String	language		

USER_ROLE			
Long	user_id	FK	
Long	role_id	FK	

PROJECT			
Long	id	PK	
String	title		
String	description		
String	thumbnail_url		
String	live_demo_url		
String	source_code_url		
Long	client_id	FK	Optional
LocalDateTime	created_at		

SKILL			
Long	id	PK	
String	name		
String	category		e.g., Frontend, Backend, DevOps
Integer	proficiency_level		0-100
String	icon_url		

experiences

sends

makes

configures

has assigned to

commissioned

utilizes